Lab Manual – 5

Code/Course: 23CSE312/Distributed Systems

Topic: Remote Procedure Call (RPC)

[CO05] [BTL-3]
Student Name: D Ganeswara
Roll No.: AM.SC.U4CSE23119
Batch/Section: CSE-B
Date:11/03/2026

Aim

To implement a simple Remote Procedure Call (RPC) service using rpcgen for an **addition** operation, compile server and client stubs, and execute a client–server RPC interaction over TCP.

add.x:

```
struct addin {
    long arg1;
    long arg2;
};

typedef long addout;

program ADD_PROG {
    version ADD_VERS {
        addout ADD_PROC(addin) = 1;
    } = 1;
} = 0x3543000;
```

Serverproc.c:

```
#include "add.h"
```

```
addout *add_proc_1_svc(addin *in, struct svc_req *rqstp)
{
    static addout result;

    result = in->arg1 + in->arg2;

    return &result;
}
```

Client.c

```
{
    CLIENT *cl;
    addin in;
    addout *out;

    if (argc != 4) {
        printf("Usage: client <machine> <int1> <int2>\n");
        return 1;
    }

    /* Create RPC client */
    cl = clnt_create(argv[1], ADD_PROG, ADD_VERS, "tcp");
    if (cl == NULL) {
        clnt_pcreateerror(argv[1]);
        exit(1);
    }

    /* Set input values */
```

```
    in.arg1 = atol(argv[2]);
    in.arg2 = atol(argv[3]);

    /* Call remote procedure */
    out = add_proc_1(&in, cl);

    if (out == NULL) {
        clnt_perror(cl, "RPC call failed");
    } else {
        printf("We received the result: %ld\n", *out);
    }

    /* Destroy client */
    clnt_destroy(cl);

    return 0;
}
```

```
root@e3f315ec8ecb:~/labsheet# ./server
```